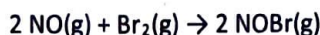


Unit 5: AP Free Response Practice #3 [8 points]

KEY #3



1. The following results were obtained in experiments designed to study the rate of the reaction above.

Experiment	Initial Concentration (mol/L)		Initial Rate of Appearance of NOBr (M/sec)
	[NO]	[Br ₂]	
1	0.02	0.02	9.6×10^{-2}
2	0.04	0.02	3.8×10^{-1}
3	0.02	0.04	1.9×10^{-1}

$$\times 4 = 2^2 \quad n=2$$

$$\times 2 = 2^1 \quad n=1$$

- a. Write the rate law for the reaction. Justify. [3 points]

$$R = k[\text{NO}]^m[\text{Br}_2]^n \quad \leftarrow \text{see above}$$

$$R = k[\text{NO}]^2[\text{Br}_2]$$

- b. Calculate the value of the rate constant, k , for the reaction. Include units. [2 points]

From Ex ①: $R = k[\text{NO}]^2[\text{Br}_2]$
 $9.6 \times 10^{-2} = k(0.02)^2(0.02)$

$$k = 1.2 \times 10^4 \text{ mol}^{-2} \cdot \text{sec}$$

- c. Determine the initial rate of the reaction in experiment 1. [1 point]

$$R = +\frac{1}{2} \frac{\Delta[\text{NOBr}]}{\Delta t} = +\frac{1}{2} (9.6 \times 10^{-2}) = 4.8 \times 10^{-2} \frac{\text{M}}{\text{sec}}$$

- d. Which of the following reaction mechanisms is consistent with the rate law established in (a)? Explain your choice. [2 points]

$$\text{II: } R = k[\text{Br}_2]$$

NOT consistent w/ Rate law!



→ I) Rate according to RDS:

$$R = k[\text{N}_2\text{O}_2][\text{Br}_2]$$

but N_2O_2 is intermediate!

$$R = \left(k \frac{k_f}{k_r}\right) [\text{NO}]^2 [\text{Br}_2]$$

$$R = k' [\text{NO}]^2 [\text{Br}_2]$$

Yes! Rate law matches for step I

From ① $R_f = R_r$

$$k_f [\text{NO}]^2 = k_r [\text{N}_2\text{O}_2]$$

$$[\text{N}_2\text{O}_2] = \frac{k_f}{k_r} [\text{NO}]^2$$

Substitute in for $[\text{N}_2\text{O}_2]$